- **A new organization provides seventy percent narrative notes and thirty percent tables with undocumented local codes. What do you do during the first two weeks?**

First, I would understand the data, required clinical concepts, and expected output.

Then I would:

- **Profile the narrative notes and tables.**
- **Identify undocumented local codes.**
- **Define a small canonical schema and mapping rules.**
- **Create a small reviewed sample with clinical/subject-matter experts.**
- **Build a small end-to-end prototype.**
- **Measure extraction accuracy and identify errors.**

I would use rules, pretrained NLP models, and potentially an LLM for exploration and labeling support. I would not treat LLM output as ground truth. Ambiguous clinical mappings would go to human review.

- **Which mappings or extractions require clinical or hospital subject-matter approval, and which can engineering approve alone?**

I think clinical approval is needed when the mapping can change clinical meaning, such as:

Diagnoses and symptoms ,Medications , Procedures, Negation and uncertainty, Clinical severity, Undocumented local codes , Conflicts between different sources and ets.
Engineering can handle technical tasks such as parsing, formatting, schema validation, timestamps, duplicates, pipeline infrastructure…

In short: engineering can own the mechanism; clinical experts should approve the clinical meaning.

- **What information should an LLM never infer when the source record does not provide it?**

An LLM should never create clinical information that is not supported by the source. For example, it should not invent: Diagnoses or symptoms, Medications, Lab values, Timing of events, Clinical severity, Causality, Patient characteristics

If information is missing, the output should remain unknown or not documented. As part of the assignment, I was asked to…

- **How do you distinguish an extraction confidence score from evidence quality and from clinical importance?**

I think these three are related, but they should remain separate.

Extraction confidence: How confident the system is that it extracted the information correctly.

Evidence quality: How strong and reliable the source information is.

Clinical importance: How important it is if the extracted information is wrong.

For example, if the source data is incomplete or ambiguous, the evidence quality is low. This may also make the extraction less reliable. However, I would not automatically use one score to calculate the others.

I would first check the quality of the available evidence and then evaluate how confidently the system extracted the information from it.

Clinical importance should affect the decision threshold. For example, if a diagnosis is much more clinically important than an administrative field, I would use a stricter threshold and send more uncertain cases for human review.

This is similar to risk management: the cost of an error is different for different clinical concepts. For a high-risk diagnosis, I would put more weight on avoiding false negatives and require stronger evidence before accepting the result.

So, I would keep confidence, evidence quality, and clinical importance as separate fields, but use them together when deciding whether to accept an extraction, reject it, or send it for review.

- **How does your design scale from one organization to twenty without creating twenty separate pipelines?**

I would use one common pipeline with organization-specific configuration, rather than twenty separate pipelines.

Each organization can have its own: Local code mappings Field names, Abbreviations ,Validation rules and …

The core code stays shared, while organization-specific differences are stored in version-controlled configuration files.